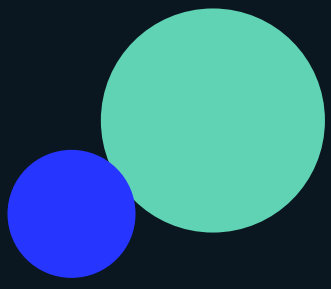
PWEB / PRODUCT BLUEPRINT

From Vision to Execution

A unified guide to the platform, its users, functionality, development progress, architecture and launch plan.

PRODUCT VISION • DEVELOPMENT RECORD • IMPLEMENTATION ROADMAP

Consolidated edition | 19 October 2026



Contents

01 Executive overview

02 The Pweb vision and product boundaries

03 Users and the end-to-end experience

04 The eight Admin modules

05 What has been built and what remains

06 Platform architecture and data model

07 Booking engine and business rules

08 Security, privacy and reliability

09 Design system and dark-mode CSS lessons

10 Roadmap, testing and release gates

11 Risks, decisions and next actions

DOCUMENT POSITIONING

This document combines and redesigns the earlier development update and vision-to-execution guides. “Prototype” refers to interface and exploratory implementation work; “planned” refers to functionality not verified as production-ready. The plan is a proposal, not a statement that a live healthcare system has been deployed.

01 Executive overview

Pweb is envisioned as one configurable, multi-institution platform for surgical practices and hospitals. Each institution would operate its own branded public-facing experience and protected internal workspace, while the Pweb platform provides shared software capabilities for scheduling, booking administration, communication and reporting.

The product in one sentence

Pweb helps practices organise appointments and specialist availability while giving patients a simpler way to request care.

Product pillars

Pillar	What it means
Institution-first	Hospital owners and practice managers are the primary purchasers; patients and staff are the daily users.
Multi-tenant	One platform serves separate institutions, with server-enforced data boundaries.
Role-based	Patients, reception staff, specialists and managers have different workflows and permissions.
Operations-led	Scheduling and appointment administration come before broader clinical records.
Incremental delivery	Prototype interfaces first; validate with a practice; build secure backend; pilot before scale.

Current status

The known work includes a hospital marketing concept, fictional hospital demonstration, responsive interface experiments and eight Admin page designs. Live booking transactions, protected patient records, robust tenant isolation and production operations have not been established by the supplied material.

02 The Pweb vision and boundaries

Who Pweb serves

Audience	Primary value
Institution owner / manager	Oversee bookings, staff access, service performance and operational settings.
Reception / administration	Capture enquiries, schedule visits, confirm requests and manage arrivals.
Specialist	Review assigned calendar, set availability and block time.
Patient	Find an institution or service, request a suitable slot and receive updates.
Pweb platform operator	Provision workspaces, manage subscriptions, monitor reliability and support.

A distinct identity for each hospital

The public website should foreground the hospital’s own name, specialties and services. Pweb remains the underlying technology brand. Each hospital could configure profile details, location, opening hours, appointment types and staff permissions.

What the first product is not

The first release should not be assumed to be an electronic health record. Clinical notes, diagnoses, prescriptions, imaging, medical billing codes and medical-aid integrations are separate high-stakes product decisions requiring deeper compliance and domain expertise.

Questions requiring customer discovery

- Is the tenant one institution or one branch? Can a hospital have multiple branches?
- Which bookings require approval rather than immediate confirmation?
- What minimum patient data is necessary to arrange an appointment?
- What should be charged per institution, branch, staff seat or booking volume?

03 End-to-end functionality

Institution setup

A verified organisation configures its workspace, services, branches, opening hours, staff and specialists. Managers define appointment durations, cancellation rules, availability and notification templates.

Patient booking journey

Step	Experience
01 — Discover	Visit the hospital-branded site and browse specialties or specialists.
02 — Find time	Select a service and see eligible availability.
03 — Request	Enter minimum contact details and consent information.
04 — Confirm	Receive an acknowledgement; admin or policy confirms where required.
05 — Attend	Reception records arrival; specialist sees the authorised schedule.
06 — Follow up	Reminders, changes, cancellations and completion are recorded.

Reception and specialist workflows

• Reception creates phone or walk-in bookings, searches patients, resolves conflicts and updates appointment states.

• Specialists view assigned appointments, manage working hours and leave, and block unavailable time.

• Managers adjust service policies, staff roles and reporting filters.

• Significant actions produce an audit record for accountability.

Appointment state model

Requested → Confirmed → Checked-in → Completed. Alternate paths include declined, cancelled and no-show. The system must enforce permitted transitions and preserve historical changes.

04 The eight Admin modules

Admin area	Core functions	Backend milestone
Home	KPI cards, activity overview, quick actions	Real-time metrics from approved data
Calendar	Day/week/month views, events, filters, dialogs	Synchronise with server-side bookings
Bookings	Requests, states, filters, details	Conflict-safe create and reschedule
Patients	Profiles, contact details, booking history	Tenant-protected records and search
Specialists	Profiles, shifts, availability, leave	Availability rules and role checks
Messages	Templates, inbox/composer, reminders	Delivery provider, queue and logs
Reports	Bookings, utilisation, cancellations	Reconciled reporting queries
Settings	Organisation, staff, appearance, permissions	Persisted organisation configuration

Visual quality requirements

Every module should remain readable in light and dark mode. Tables need clear row and header contrast; calendar cells must distinguish selected, outside-month and unavailable states; status badges must remain meaningful; forms, dialogs, hover states and keyboard focus must be visible on mobile and desktop.

05 Development record and current state

Progress represented in the earlier guides

• Product positioning shifted from a general Pweb portfolio concept toward hospital and surgical-practice operations.

• A fictional hospital visitor experience and hospital-facing marketing/carousel ideas were explored.

• The Admin information architecture includes eight distinct sections with shared navigation.

• Responsive laptop/phone mockups, mobile navigation and cross-page relative asset paths were addressed.

• Light, Dark and System appearance settings were introduced, with a shared JavaScript preference and standalone dark-mode CSS.

Implementation status

Area	Status / interpretation
Frontend UI	Designed or prototyped; ongoing responsive and dark-mode QA.
Theme preference	Browser preference approach established using localStorage and system setting.
Appointments	Interface work exists; real transactional API not verified.
Authentication and roles	Planned; production enforcement not verified.
Tenant isolation	Required design decision; not verified in live backend.
Notifications, billing, reports	Proposed integrations and services; not verified production functionality.

Immediate frontend milestone

Audit all eight Admin pages on desktop, tablet and mobile; ensure no white-on-white cards, unreadable table text or invisible calendar events; preserve the existing top navigation while making page content theme-aware.

06 Architecture and data

Suggested technical layers

Layer	Responsibility
Public and private frontends	Hospital visitor page, patient flow, specialist workspace and Admin UI.
Backend API	Identity, authorisation, tenant context, bookings, availability and settings.
Relational database	Institutions, branches, staff, patients, specialists, appointments and logs.
Background workers	Reminders, retries, reports, integration callbacks.
Operations	Monitoring, error logs, backups, incident response and deployment.

Core entities

Tenant (organisation), Branch, User/Membership, Specialist, Patient, Appointment, Availability Rule, Notification and Audit Event. Tenant-owned records require tenant identifiers and every relevant server query must enforce tenant scope.

Dependencies

Authentication and tenant boundaries precede live patient data. Availability rules precede bookable slots. Reliable booking events precede notifications and analytics. Audits, backup recovery and support procedures precede a real-world pilot.

07 Booking engine and operational rules

How available slots are derived

Recurring specialist hours + branch opening hours − blocked periods − leave − existing bookings − relevant resource constraints. The engine also considers appointment duration, buffers, lead time and applicable time zones.

Preventing double-booking

A slot shown in the browser is not a guarantee. The server must validate availability and commit the reservation atomically, using appropriate database constraints and transaction handling. Idempotency protects against duplicate requests caused by retries.

Acceptance tests

- Two simultaneous requests cannot both confirm the same exclusive specialist slot.
- A cancellation releases capacity only according to policy.
- Rescheduling updates reminders while preserving the prior event history.
- An unauthorised staff member cannot override blocked hours.
- A patient or staff member cannot retrieve another institution’s protected records.

Messages and reporting

Confirmed events should enqueue email reminders and track delivery state. Reporting should use defined metrics: requested, confirmed, completed, cancelled and no-show appointments; lead time; specialist utilisation; and branch demand. Avoid showing invented or unreconciled numbers in production dashboards.

08 Security, privacy and reliability

Essential controls

• Use well-tested authentication, secure sessions, strong password storage and MFA for privileged accounts.

• Check role and tenant permissions at every protected endpoint and record access.

• Encrypt traffic with TLS; encrypt stored sensitive data with managed keys and restricted access.

• Validate inputs; apply appropriate CSRF protection, rate limits, security headers and secret management.

• Maintain tamper-resistant audit logs, backups, restore tests and dependency updates.

• Use minimal patient data and define retention, deletion and incident-response procedures.

South African healthcare context

Before processing real patient information, obtain professional guidance about POPIA, healthcare confidentiality, recordkeeping, processor/operator agreements, breach procedures and any cross-border data transfers. This guide is not legal advice or a compliance certification.

Operational readiness

Define support contacts, service availability targets, incident severity, backup recovery-point objectives and recovery-time objectives. A secure-looking interface is not evidence of a secure or reliable backend.

09 Design system and CSS engineering

Pweb visual identity

Token	Application
#2635FF — Pweb blue	Primary brand, key headings, links and active accents.
#0A1620 — Deep navy	Cover, navigation identity and high-contrast dark surfaces.
#5FD3B3 — Teal	Secondary accent, highlights and accessible success cues.
Light slate / white	Day-mode page backgrounds, cards and whitespace.
Blue-grey dark surfaces	Night-mode cards, inputs, tables and dialogs.

Why some dark-mode cards remained white

The original Admin CSS contains nested rules such as .ab-page .ab-stat { background: #fff; }. An override fails when it targets the wrong element, loses due to the cascade, or is loaded incorrectly. For normal author declarations, unlayered rules outrank layered rules; for !important rules, layer precedence reverses. Specificity is considered after origin, importance and layer precedence.

Recommended approach

• Keep one shared pweb-dark-mode.css loaded after each page’s stylesheets, and one pweb-theme.js for preference application.

• Use html[data-theme="dark"] with page-specific selectors that match the real DOM; keep normal dark overrides unlayered when appropriate.

• Avoid generic selectors that unintentionally change top navigation, SVG icons or unrelated pages.

• Inspect the Computed panel for a stubborn white card to identify the winning background declaration.

• Test tables, calendar day cells, status tags, form fields, modals, hover and focus states in all three theme settings.

Path and preference conventions

For HTML under pages/, links to root assets typically use ../. Store UI theme choice under pweb_theme in localStorage; use system colour preference when System is selected. Never use localStorage as a secure patient-data store.

10 Roadmap and delivery gates

Phase	Deliverable	Exit gate
0 — Discovery	Interview practices, map workflows, MVP scope	Validated requirements
1 — Foundation	Project structure, auth, tenant data model	Tenant isolation tests pass
2 — Booking core	Specialist schedules, booking API, calendar	No double-booking in tests
3 — Operations	Patients, notifications, audit, settings	Simulated full-day workflow
4 — Pilot	One practice, training, security and restore tests	Pilot feedback and acceptance
5 — Commercial	Subscription, support, monitoring	Controlled paid launch
6 — Expansion	Multi-branch, integrations, advanced analytics	Demand-driven prioritisation

Illustrative 12-week prototype plan

Weeks 1–2: requirements, schema and threat model. 3–4: organisations, authentication, roles and specialists. 5–7: availability, booking transactions and calendar. 8–9: patient operations, notifications and audit. 10–11: reports, accessibility and automated QA. 12: simulated pilot, security review and recovery exercise. This is a planning example, not a guaranteed schedule.

Testing layers

Unit tests for slot calculations and state transitions; integration tests for bookings and delivery; security tests for tenant/role bypass; responsive UI checks for light/dark/system; operational tests for backups, deployment rollback and message retries.

11 Risks, decisions and next actions

Risk	Mitigation
Patient information exposure	Minimise collection; authorise on server; audit access.
Double-booked appointments	Transactional writes and database constraints.
Too many features	Freeze a narrow MVP; defer clinical records.
Undelivered reminders	Queue, retries and delivery-state tracking.
Cross-hospital leakage	Tenant isolation in schema, queries and tests.
Difficult onboarding	Pilot with one willing practice.
Unclear commercial value	Interview buyers and test pricing assumptions.

Next five actions

- Write a one-page MVP scope with explicit exclusions.
- Map three flows: patient booking, reception booking and specialist calendar.
- Choose a backend/database and define tenant membership and role permissions.
- Connect Calendar and Bookings interfaces to one secure test API.
- Ask a real practice administrator to review the flow before implementing complex billing.

Closing principle

Pweb should grow from a convincing prototype into a dependable operations service through deliberate validation, narrow releases and strong security. The first meaningful success is not a large feature list: it is one institution completing a full appointment workflow accurately and safely.

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